



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Master of Engineering – M.E in Computer Science and Engineering

Course Plan

Course Name	:	Applied Linear Algebra
Course Code	:	AME 5101
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Dr. Devi Prasad M
Name of the Program Coordinator	:	Dr. Sathyendranath Malli

Date:	10/8/2026
Signature of Program Coordinator	
with Date	

Date:	10/8/2026
Signature of Course Coordinator	
with Date	



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Table of Contents	
1 Course Plan	5
1.1 Primary Information	5
1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping	6
1.3 Assessment Plan	7
1.4 Lesson Plan	8
1.5 References	11
1.6 Other Resources (Online, Text, Multimedia, etc.)	11
1.7 Course Timetable	12
1.8 Assessment Plan	13
1.9 Assessment Details	15
1.10 Course Articulation Matrix	16



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering – M.E in Computer Science and Engineering, programme are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in computer science.
PEO 2	Introduce state of art technologies in computer science and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering – M.E in Computer Science and Engineering, graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop software based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analysing and decision-making.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1. Course Plan

1.1 Primary Information

Course Name	:	Applied Linear Algebra [AME 5101]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Basic Algebra and Calculus
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Apply vector operations for similarity learning.	10	PO3	3
CO2	Analyse matrix representations in AI & ML models.	14	PO4,PO3	4
CO3	Apply matrix calculus for optimization in learning systems.	12	PO3	3



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	2 assignments. 15 days each.	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Applying and Analyzing	Applying and Analyzing	Applying and Analyzing
Pattern	Answer all 5 questions of 10 marks each. Each question may have 2 parts of equal marks.	Apply and analyze solutions to real-world problems.	Answer all 10 questions of 10 marks each. Each question may have 2 parts of equal marks.
Schedule	As per academic calendar.	September 15 and October 10	As per academic calendar.
Topics covered	Vectors, matrices, GPU/TPU support for deep learning systems.	Vectors, matrices, efficient solutions on GPU/TPUs.	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Conceptual introduction to vector spaces and vector operations. Scalar product.	C01
L2	Introduction to matrices. Types of matrices. Operations. Matric-vector multiplication.	C02
L3	Inner and outer products. Norm of a vector. Distance - standard deviation. Standardizations versus normalization.	C01
L4	Linear independence and basis vectors.	C02
L5	High-dimensional vector embeddings as in Word2Vec and LLMs.	C02
L6	Linear operations on embeddings: Finding relationships, combining and isolating concepts, and analogies.	C01
L7	Systems of linear equations: over and under-determined systems.	C01
L8	Gram-Schmidt algorithm.	C03
L9	QR-factorization.	C02
L10	Solving linear equations.	C01
L11	Applications in deep learning - Introduction to sensitivity and gradient in one dimension.	C03



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

L12	Chain rule. Matrix derivatives.	C03
L13	Gradient descent.	C02
L14	Training the artificial neural networks - backward propagation of errors.	C02
L15	Backpropagation algorithm.	C03
L16	Introduction to Attention mechanism.	C01
L17	Implementation details of attention mechanism in large language models.	C03
L18	Implementation details of attention mechanism in large language models.	C03
L19	Tensor cores and GPU architectures	C02
L20	Tensor cores and GPU architectures	C02
L21	Least squares: problem motivation and examples.	C02
L22	Solving least squares problems - least square data fitting.	C01
L23	Solving least squares problems - least square data fitting.	C03
L24	Maximizing variance and minimizing projection error.	C01
L25	Minimizing projection error, and identifying principal components visually.	C03



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

L26	Mean centering and covariance matrices.	C02
L27	Covariance matrices, eigenvectors, and eigenvalues.	C03
L28	Mean centering, covariance matrices, eigenvectors, and eigenvalues - Data Preparation and Relations.	C03
L29	Mean centering, covariance matrices, eigenvectors, and eigenvalues - The Transformation Engine.	C03
L30	Feature scaling (standardization), eigendecomposition vs. Singular Value Decomposition (SVD)	C03
L31	Applying PCA to solve machine learning bottlenecks	C01
L32	Applying PCA to solve machine learning bottlenecks	C01
L33	Loss of feature interpretability, sensitivity to outliers, and the limitation of linear assumptions	C02
L34	Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-batch Gradient Descent. Vectorization efficiency, memory constraints, convergence speed, and escaping local minima.	C02
L35	Sigmoid with Binary Cross-Entropy vs. Softmax with Categorical Cross-Entropy. Mapping out the backward pass differences between two-class and many-class problems.	C02
L36	Linear/Identity activation functions at the output layer and Mean Squared Error (MSE) loss. Comparing gradient behavior and scale stability between regression and classification losses	C02



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1.5 References

1. Introduction to Applied Linear Algebra, Vectors, Matrices, and Least Squares, Stephen Boyd & Lieven Vandenberghe, Cambridge University Press, 1st Edition, 2018. Available online at <http://vmls-book.stanford.edu/vmls.pdf>.
2. Linear Algebra and Learning from Data, Gilbert Strang, Cambridge Uni. Press, 1st Edition, 2019.
3. IBM AI Engineering Professional Certificate (<https://www.coursera.org/professional-certificates/ai-engineer>).
4. Deep Learning with Python, François Chollet, Manning Publications, 2nd Edition, 2021.
5. Neural Networks and Deep Learning, Michael A. Nielsen, Determination Press, 2015. Available online at <http://neuralnetworksanddeeplearning.com/>
6. Matrix Methods: Applied Linear Algebra, Richard Bronson and Gabriel B. Costa, Academic Press, 3rd Edition, 2008.
7. Introduction to computational thinking (<https://computationalthinking.mit.edu/Fall24/>).

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



1.7 Course Timetable

I Semester						Room: LG01-LH08			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday		ALA							
Tuesday									
Wednesday		ALA							
Thursday									
Friday		ALA							
Saturday									



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1.8 Assessment Plan

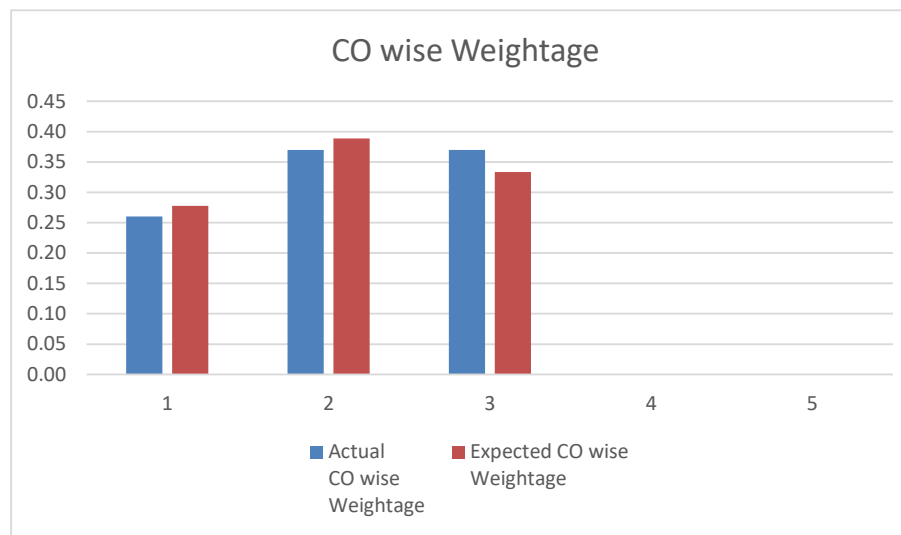
COs		Marks & Weightage				
CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Apply vector operations for similarity learning.	10	10	20	0.26	0.28
C02	Analyse matrix representations in AI & ML models.	20	5	40	0.37	0.39
C03	Apply matrix calculus for optimization in learning systems.	20	5	40	0.37	0.33
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)



Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

$$\text{Weightage for CO1} = (\text{MSE marks for CO1} / 1.6666 + \text{Assignment marks for CO1} / 1.0 + \text{ESE marks for CO1} / 2) / 100$$



MANIPAL SCHOOL OF INFORMATION SCIENCES

MANIPAL

(A constituent unit of MAHE, Manipal)

1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	- Performance is measured using MSE attainment level. - Reference: question paper and answer scheme. - MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	- Performance is measured using assignments/quiz attainment level. - Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	- Performance is measured using ESE attainment level. - Reference: question paper and answer scheme. - ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2			Y	Y	
CO3			Y		
Average Articulation Level			*	*	